

ZK-PoC device probe

Measures achievable F(d) – GFLOPS through the browser – on the WebGPU and CPU paths, at a controlled duty cycle. Output feeds DEFAULT_TIERS in bench/breakeven.py, replacing the literature-anchored placeholders.

1 • Device

user agent Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36

logical cores	8
device memory	16 GB (coarse)
WebGPU	available
GPU adapter	intel / gen-12lp
battery	34% (discharging)

2 • Benchmark

matrix N

256

 duty cycle

100% (peak F(d))

 reps

7

Run probe

Copy JSON

done

3 • Results

path	achievable F(d)	median	check
CPU (JS typed array)	0.59 GFLOPS	57.3 ms	–
GPU (WebGPU / WGSL)	7.80 GFLOPS	4.3 ms	verified (rel err 0.0e+0)
config	N=256, duty=100%, 2N³=0.03 GFLOP/rep		
battery drop	0.00 % over the run		

4 • Export

```
{
  "schema": "zkpoc.device-probe/1",
  "captured_at": "2026-08-03T05:55:24.653Z",
  "config": {
    "matrix_n": 256,
    "duty_cycle": 1,
    "reps": 7
  }
}
```

```
},
"device": {
  "ua": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/1
  "platform": "Win32",
  "cores": 8,
  "memory_gb": 16,
  "webgpu": true,
  "adapter": {
    "vendor": "intel",
    "architecture": "gen-12lp",
    "device": null,
    "description": null
  },
  "battery": {
    "level": 0.34,
    "charging": false,
    "dischargingTime": 2730
  },
  "limits": {
    "maxComputeWorkgroupStorageSize": 32768,
    "maxBufferSize": 2147483648
  }
},
"battery_after": {
  "level": 0.34,
  "charging": false,
  "dischargingTime": 2730
},
"results": {
  "cpu": {
    "path": "js-typed-array",
    "median_ms": 57.30000000447035,
    "min_ms": 45.399999998509884,
    "max_ms": 102.60000000149012,
    "samples": [
      45.399999998509884,
      48.100000001490116,
      53.19999999552965,
      57.30000000447035,
      58.30000000447035,
      94.5,
      102.60000000149012
    ],
    "gflops": 0.5855921814551868
  },
  "gpu": {
    "path": "webgpu-wgsl",
    "median_ms": 4.299999997019768,
    "min_ms": 3.2000000029802322,
    "max_ms": 7,
    "samples": [
      3.2000000029802322,
      3.2999999970197678,
      3.7999999970197678,
      4.299999997019768,
      4.600000001490116,
```

```
    5.0999999940395355,  
    7  
  ],  
  "gflops": 7.803356284478097,  
  "correct": true,  
  "rel_error": 0  
}  
},  
"note": "cpu row is a JS lower bound, not WASM SIMD; see M1"
```

Scope. The CPU row uses a JS typed-array kernel, which is a defensible lower bound but not the Rust→WASM SIMD path – that kernel lands in M1 and will replace this row. The WebGPU row is the one that matters: the break-even model shows the CPU path is economically dead regardless.

Energy. Browsers expose no power draw. Battery Status is sampled where available, but real W(d) needs host-side RAPL / Intel Power Gadget / powermetrics – see [bench/power/](#). Quantifying energy and thermal effects is the gap FibRace (arXiv:2510.14693 §5.1.3) explicitly left open, so it is instrumented deliberately rather than assumed.